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Quadratic Equations

Grade 11 • Mathematics • Module 2

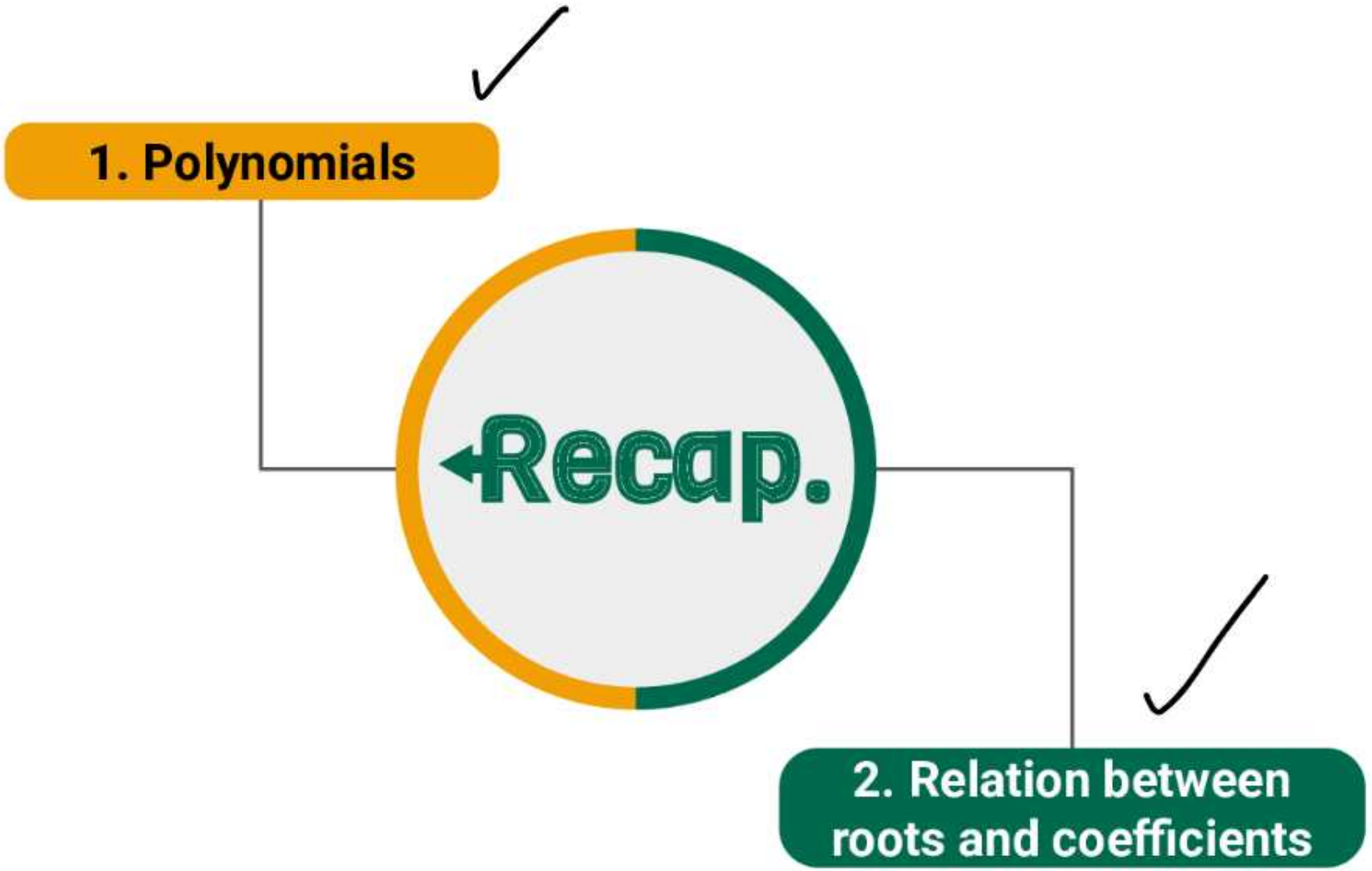




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Recap.





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Quiz.



Quiz.

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Q. If the product of roots of the equation $(a+1)x^2+(2a+3)x+4=0$ is 2 then the sum of roots is

A $\frac{5}{2}$

B $\frac{-5}{2}$

C $\frac{3}{2}$

D $\frac{1}{2}$



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Q. If the product of roots of the equation $(a+1)x^2+(2a+3)x+4=0$ is 2 then the sum of roots is

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$S_2 = \alpha\beta = c/a$

$S_1 = \alpha + \beta = -\frac{b}{a}$

Q. If the product of roots of the equation $(a+1)x^2 + (2a+3)x + 4 = 0$ is 2 then the sum of roots is

Solution

Given, $(a+1)x^2 + (2a+3)x + 4 = 0$

$$\alpha\beta = \frac{c}{a} = \frac{4}{a+1} = 2$$

$$4 = 2a + 2$$

$$2a = 2$$

$$a = 1$$

$$\alpha + \beta = \frac{-b}{a} = \frac{-(2a+3)}{a+1}$$

$$\alpha + \beta = \frac{-5}{2}$$

$S_2 = \frac{c}{a} = 2$ (1)

$c = 2a$

$4 = 2(a+1)$

$4 = 2a + 2$

$2a = 2$

$a = 1$

$S_1 = \alpha + \beta = -\frac{b}{a}$

$= -\frac{(2a+3)}{a+1}$

$= -\left(\frac{5}{2}\right)$

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1 Polynomial

2 **Relation between Roots and Coefficients** Continuing

3 **Nature of roots**

4 Transformation of Equations

5 Common Roots

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Contents

1 Polynomial

2 **Relation between Roots and Coefficients**

3 Nature of roots

4 Transformation of Equations

5 Common Roots

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2. Relation Between Roots and Coefficients

For a cubic equation, we have

$$ax^3 + bx^2 + cx + d = 0,\ a \neq 0,\ a, b, c, d \in \mathbb{R}$$

$$S_1 = \alpha + \beta + \gamma = -\frac{b}{a}$$

$$S_2 = \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$$

$$P = \alpha\beta\gamma = -\frac{d}{a}$$

n = degree

$$S_n = (-1)^n \frac{\text{Coeff of } x^{n-n}}{\text{Coeff of } x^n}$$

n=3

$$S_1 = (-1)^1 \frac{\text{Coeff of } x^{3-1}}{\text{Coeff of } x^3} = -\frac{\text{Coeff of } x^2}{\text{Coeff of } x^3}$$

n=1

$$= -b/a$$

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2. Relation Between Roots and Coefficients

For a biquadratic equation, we have

$$ax^4 + bx^3 + cx^2 + dx + e = 0,\ a \neq 0,\ a, b, c, d, e \in \mathbb{R}$$

$$S_1 = \alpha + \beta + \gamma + \delta = -\frac{b}{a}$$

$$S_2 = \alpha\beta + \beta\gamma + \gamma\delta + \alpha\gamma + \alpha\delta + \beta\delta = \frac{c}{a}$$

$$S_3 = \alpha\beta\gamma + \beta\gamma\delta + \gamma\delta\alpha + \alpha\beta\delta = \frac{-d}{a}$$

$$P = \alpha\beta\gamma\delta = \frac{e}{a}$$

$n = 4$

$$S_n = (-1)^n \frac{\text{Coeff of } x^{n-n}}{\text{Coeff of } x^n}$$

$$S_1 = (-1)^1 \frac{\text{Coeff of } x^{4-1}}{\text{Coeff of } x^4}$$

$$= -\frac{b}{a}$$

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$$ax^5 + bx^4 + cx^3 + dx^2 + ex + f = 0$$

$\alpha, \beta, \gamma, \delta, \epsilon$

$S_1 = \frac{-b}{a} = \sum \alpha$

$S_2 = +\frac{c}{a} = \sum \alpha\beta$

$S_3 = -\frac{d}{a} = \sum \alpha\beta\gamma$

$S_4 = +\frac{e}{a} = \sum \alpha\beta\gamma\delta$

$S_5 = -\frac{f}{a} = \alpha\beta\gamma\delta\epsilon$

$S_n = (-1)^n \frac{\text{coeff of } x^{n-n}}{\text{coeff of } x^n}$

$n = 5$

$S_2 = \alpha(\beta+\gamma+\delta+\epsilon) + \beta(\gamma+\delta+\epsilon) + \gamma(\delta+\epsilon) + \delta(\epsilon)$



Let's Solve

a=3 b=-5 c=-11 d=-3

Ex. If $p(x) = 3x^3 - 5x^2 - 11x - 3$ is a polynomial of degree three then write the relationship between zeros and the coefficients.

roots

Solution

Given, $3x^3 - 5x^2 - 11x - 3$ is polynomial of degree three and it is from

$$ax^3 + bx^2 + cx + d = 0$$

$\Sigma\alpha$: $\alpha + \beta + \gamma = \dfrac{-b}{a} = -\dfrac{(-5)}{3}$

$\Sigma\alpha\beta$: $\alpha\beta + \beta\gamma + \gamma\alpha = \dfrac{c}{a} = -\dfrac{11}{3}$

$\alpha\beta\gamma = -\dfrac{d}{a} = \dfrac{3}{3} = 1$

$S_1 = -\dfrac{b}{a} = -\dfrac{(-5)}{3} = 5/3$

$S_2 = +\dfrac{c}{a} = -\dfrac{11}{3}$

$S_3 = -\dfrac{d}{a} = -\dfrac{(-3)}{3} = 1$

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Let's Solve



Let's Solve

Ex. If $p(x) = 3x^3 - 5x^2 - 11x - 3$ is a polynomial of degree three then write the relationship between zeros and the coefficients.

✓

$$S_1 = -\frac{b}{a}$$

$$S_2 = +\frac{c}{a}$$

$$S_3 = -\frac{d}{a}$$

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Let's Solve

Ex. If $p(x)=3x^3-5x^2-11x-3$ is a polynomial of degree three then write the relationship between zeros and the coefficients.

Solution

Given, $3x^3-5x^2-11x-3$ is polynomial of degree three and it is from

$$ax^3+bx^2+cx+d=0$$

$$\alpha+\beta+\gamma=\frac{-b}{a}=-\frac{(-5)}{3}$$

$$\alpha\beta+\beta\gamma+\gamma\alpha=\frac{c}{a}=-\frac{11}{3}$$

$$\alpha\beta\gamma=-\frac{d}{a}=\frac{3}{3}=1$$

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Quiz.



Quiz.

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Q. If α, β and γ are the roots of $x^3 - 2x + 1 = 0$ then find sum of roots:

A 0 **B** 2 **C** -2 **D** 1



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$x^3 \quad x^2 \quad x$

$a=1 \quad b=0 \quad c=-2 \quad d=1$

Q. If α, β and γ are the roots of $x^3 - 2x + 1 = 0$ then find sum of roots:

A 0 **B** 2 **C** - 2 **D** 1

$S_1 = \alpha + \beta + \gamma = -\frac{b}{a} = -\frac{0}{1} = 0$

$S_2 = \alpha\beta + \beta\gamma + \gamma\alpha = +\frac{c}{a} =$

(P) $S_3 = \alpha\beta\gamma = -\frac{d}{a}$

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$a=1\ b=0\ c=-2\ d=-1$

Q. If α, β and γ are the roots of $x^3-2x+1=0$ then find sum of roots:

$ax^3+bx^2+cx+d=0$

Solution

Given, α, β and γ are the roots of $x^3-2x+1=0$

Thus, $\alpha+\beta+\gamma=0$

$S_1=\frac{-b}{a}$

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Let's Solve



Let's Solve

Ex. If 2, 3 are roots of $2x^3 + mx^2 - 13x + n = 0$, then find m, n & third root.

2 roots → $2(2)^3 + m(2)^2 - 13(2) + n = 0$

$16 - 4m - 26 + n = 0$

$-4m + n - 10$ — (1)

3 root → $2(3)^3 + m(3)^2 - 13(3) + n = 0$

$54 - 9m - 39 + n = 0$

$-9m + n - 15 = 0$ — (2)

(1) − (2)

$5m + 5 = 0$

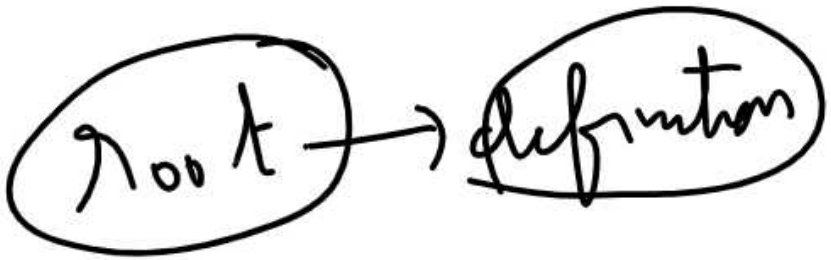
$m = -1$

$n = ?$

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Let's Solve

Ex. If 2, 3 are roots of $2x^3 + mx^2 - 13x + n = 0$, then find m, n & third root.



Solution

Method I:

Let third root be α.

$$\therefore \alpha + 2 + 3 = \frac{-m}{2} \quad \ldots(1)$$

$$2\alpha + 6 + 3\alpha = \frac{-13}{2} \quad \ldots(2)$$

$$6\alpha = \frac{-n}{2} \quad \ldots(3)$$

$$(2) \Rightarrow \alpha = \frac{-5}{2}$$

Putting in (1) we get $m = -5$

Putting in (3) we get $n = 30$

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Let's Solve

Ex. If 2, 3 are roots of $2x^3 + mx^2 - 13x + n = 0$, then find m, n & third root.

Solution

Method II:

Since 2 and 3 are root of equation

Thus, it will satisfy the equation

So $2(2)^3 + m(2)^2 - 13(2) + n = 0$

or $16 - 4m - 26 + n = 0$

$4m + n = 10 \quad .....(1)$

Again, $2(3)^3 + m(3)^2 - 13(3) + n = 0$

$54 + 9m + n = 0$

$9m + n = -15 \quad .....(2)$

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Let's Solve

Ex. Let α & β are the roots of $x^3 + px^2 + qx + r = 0$ such that $\alpha\beta = -1$ prove that either $r = 0$ or $r^2 + pr + q + 1 = 0$.

$\alpha^3 + p(\alpha)^2 + q(\alpha) + r = 0$ — (1)

$\beta^3 + p(\beta)^2 + q(\beta) + r = 0$ — (2)

$S_1 = \alpha + \beta + \gamma = -p$

$S_2 = \alpha\beta + \beta\gamma + \gamma\alpha = q$

$S_3 = \alpha\beta\gamma = -r$

$-1(\gamma) = r$

$\gamma = r$

γ root $\rightarrow$ $\gamma^3 + p(\gamma)^2 + q(\gamma) + r = 0$

$\gamma^3 + p\gamma^2 + q\gamma + r = 0$

$\gamma(\gamma^2 + p\gamma + q + 1) = 0$

Hence

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Let's Solve

Ex. Let α & β are the roots of $x^3 + px^2 + qx + r = 0$ such that $\alpha\beta = -1$ prove that either $r = 0$ or $r^2 + pr + q + 1 = 0$.

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Let's Solve

Ex. Let α & β are the roots of $x^3 + px^2 + qx + r = 0$ such that $\alpha\beta = -1$ prove that either $r = 0$ or $r^2 + pr + q + 1 = 0$.

Solution

Let roots be, α, β and γ

Thus, $\alpha + \beta + \gamma = -p$

$\alpha\beta + \beta\gamma + \alpha\gamma = q$

$\alpha\beta \cdot \gamma = -r$

Now, $\alpha\beta = 1$

thus, $(-1)\gamma = -r$

$\gamma = r$

So, 'r' is one of the root

Thus, it will satisfy the equation

$r^3 + pr^2 + qr + r = 0$

$r\left(r^2 + pr + q + 1\right) = 0$

Thus either $r = 0$

or $r^2 + pr + q + 1 = 0$

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6 Analysis of graph of Quadratic Equations

7 Location of Roots

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$ax^2 + bx + c = 0$

3. Nature of Roots

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

D

$$D = b^2 - 4ac$$

= Value
(Real Number)



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3. Nature of Roots

Value of discriminant 'D' determine the nature of roots of equation,

for a quadratic equation $ax^2 + bx + c = 0$

where $a, b, c \in R$ & $a \neq 0$

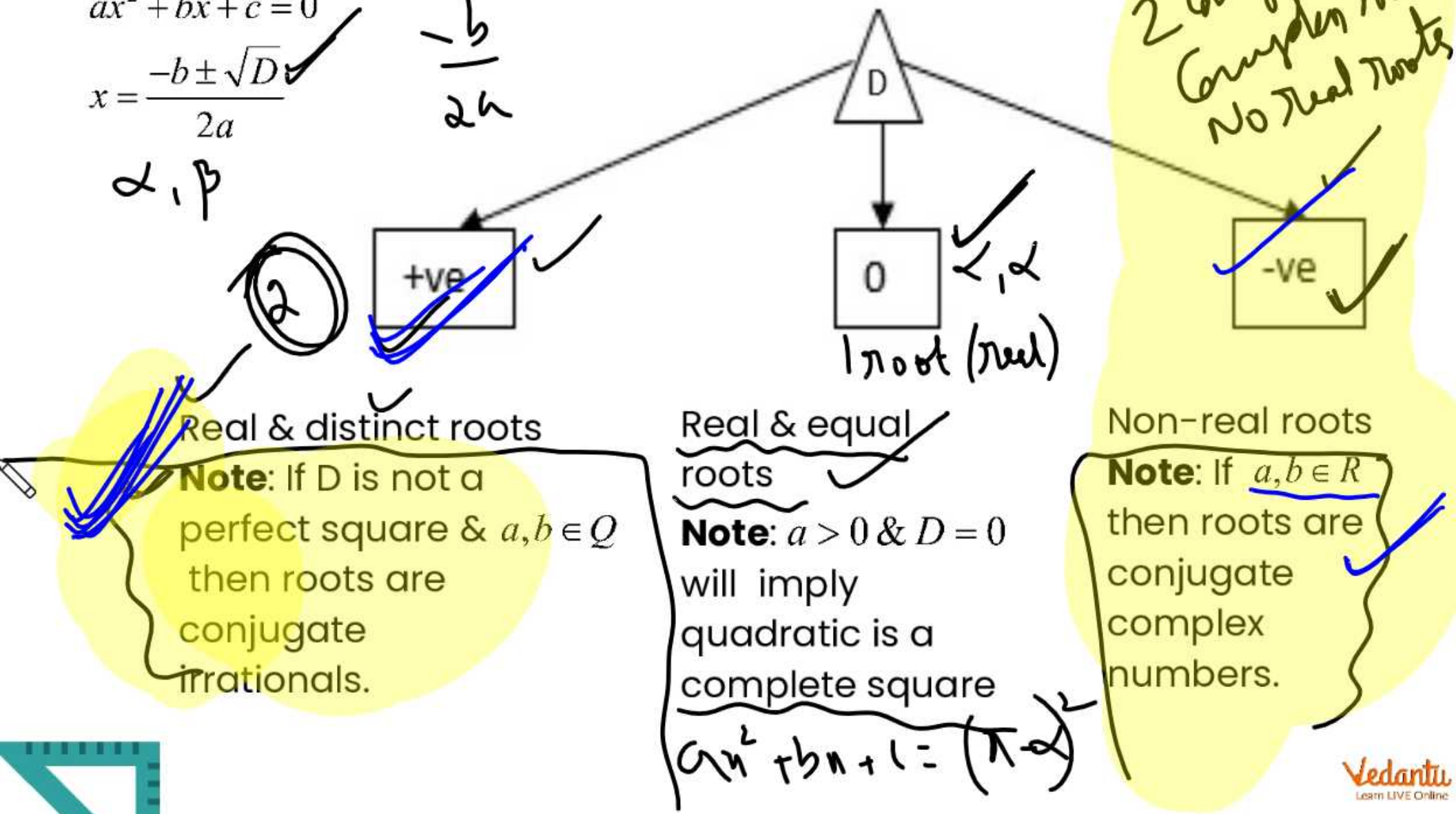
Discriminant $D = b^2 - 4ac$

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3. Nature of Roots

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{D}}{2a}$$



$ax^2+bx+c=0$

$D=b^2-4ac$

<0

Complex roots

$2+i$
$2-i$
$1+i\sqrt{3}$
$1-i\sqrt{3}$

$\alpha-iy$
$\alpha+iy$

$a,b,c \in \mathbb{R}$

$p+iq$
$p-iq$

2 Complex Conjugate

$p+ia$

$a,b,c \notin \mathbb{R}$

$a+ib+jc$ is a complex number

$l+im$

$D = b^2 - 4ac$ ✓

$x = \frac{b \pm \sqrt{b^2-4ac}}{2a}$ ✓ (b^2-4ac = D)

so

Real & distinct

D is NOT a perfect square

$C = \sqrt{2}+1$

$\frac{-2+\sqrt{3}}{4}$ ✓

$\frac{-2-\sqrt{3}}{4}$

$a, b, c \in \mathbb{Q}$

$a+\sqrt{b}$ $a-\sqrt{b}$

Conjugate/irrational

$a, b, c \notin \mathbb{Q}$

Let's Solve



Let's Solve

$D = b^2 - 4ac$

Ex. Discuss the nature of roots of the equation $6x^2 - 7x + 2 = 0$

$D = (-7)^2 - 4(6)(2)$

$49 - 48 = 1 > 0$

Solution

Comparing $6x^2 - 7x + 2 = 0$ with $ax^2 + bx + c = 0$ we have

Here, $a = 6, b = -7, c = 2$.

Therefore, discriminant = $b^2 - 4ac = (-7)^2 - 4.6.2 = 49 - 48 = 1 > 0$

Therefore, the roots are real and unequal.

real / distinct

$\neq 1, 0$

$\downarrow$

N, W, Q, Z



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Let's Solve

Ex. Discuss the nature of roots of the equation $6x^2 - 7x + 2 = 0$

Solution

Comparing $6x^2 - 7x + 2 = 0$ with $ax^2 + bx + c = 0$ we have

Here, $a = 6, b = -7, c = 2$.

Therefore, discriminant = $b^2 - 4ac = (-7)^2 - 4.6.2 = 49 - 48 = 1 > 0$

Therefore, the roots are real and unequal.

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Quiz.



Quiz.

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$D = b^2 - 4ac$

$D < 0$

$a, b, c \in \mathbb{R}$

Q. The equation $3x^2 + 4x + 6 = 0$ has

A two real roots

B one imaginary and one real

C no real roots → 2 complex

D None of the above

$D = (4)^2 - 4(3)(6)$

$16 - 72 < 0$

Conjugate

$a + ib$

$a - ib$

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Quiz.

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Q. The equation $3x^2+4x+6=0$ has

Solution

Here, $a=3, b=4, c=6.$

So, the discriminant $=b^2-4ac$

$=4^2-4.3.6=16-72=-56<0$

Therefore, the roots of the given equation are not real.

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Quiz.

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Q. The equation $3x^2+4x+6=0$ has

Solution

Here, $a=3, b=4, c=6.$

So, the discriminant $=b^2-4ac$

$=4^2-4.3.6=16-72=-56<0$

Therefore, the roots of the given equation are not real.

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Let's Solve



Let's Solve

Ex. If a, b, c are real, then find the nature of the roots of following equation $3x^2 - 2(a+b+c)x + bc + ca + ab = 0$

Solution

Given, $3x^2 - 2(a+b+c)x + bc + ca + ab = 0$

The discriminant D is given by

$$D = 4(a+b+c)^2 - 4 \times 3(bc + ca + ab)$$

$$= 4\left[a^2 + b^2 + c^2 + 2bc + 2ca + 2ab - 3bc - 3ca - 3ab\right]$$

$$= 4\left[a^2 + b^2 + c^2 - bc - ca - ab\right] = 4 \cdot \frac{1}{2}\left[(a-b)^2 + (b-c)^2 + (c-a)^2\right]$$

which is always ≥ 0

As a, b, c are real, we get $D \geq 0$,thus roots of the equation are real.

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Let's Solve

Ex. If a,b,c are real, then find the nature of the roots of following equation $3x^2-2(a+b+c)x+bc+ca+ab=0$

Solution

Given, $3x^2-2(a+b+c)x+bc+ca+ab=0$

The discriminant D is given by

$$D=4(a+b+c)^2-4\times3(bc+ca+ab)$$

$$=4\left[a^2+b^2+c^2+2bc+2ca+2ab-3bc-3ca-3ab\right]$$

$$=4\left[a^2+b^2+c^2-bc-ca-ab\right]$$

which is always ≥ 0

As a,b,c are real, we get $D\geq0$,thus roots of the equation are real.

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Let's Solve



Let's Solve

Ex. Find the value of 'p', if the roots of the following quadratic equation $(p-3)x^2+6x+9=0$ are Coincident → Equal roots

$D = 0$

$b^2 - 4ac = 0$

$(6)^2 - 4(p-3)(9) = 0$

$36 - 36(p-3) = 0$

$1 - (p-3) = 0$

$1 - p + 3 = 0$

$p = 4$

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Let's Solve

Ex. Find the value of 'p', if the roots of the following quadratic equation $(p-3)x^2+6x+9=0$ are Coincident

$D = b^2 - 4ac = 0$

Solution

For the equation $(p-3)x^2+6x+9=0$

$a = p-3, b = 6$ and $c = 9$

Since, the roots are equal

Therefore, $b^2-4ac=0$

$\Rightarrow (6)^2-4(p-3)\times 9=0$

$\Rightarrow 36-36p+108=0$

$\Rightarrow 144-36p=0$

$\Rightarrow p=\frac{144}{36}$

$\Rightarrow p=4$

Therefore, the value of $p=4$

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Quiz.



Quiz.

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Q. What is the value of 'k' if the quadratic equation $x^2 - 6x + k = 0$ has equal roots?

A 6 **B** 3 **C** 9 **D** 12

$D = 0$

$(-6)^2 - 4(1)(k) = 0$

$4k = 36$

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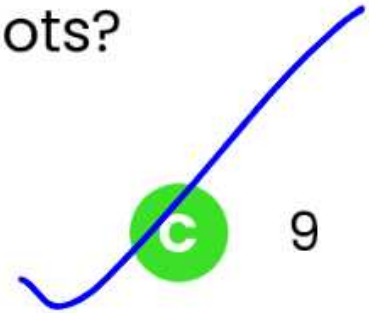
Q. What is the value of 'k' if the quadratic equation $x^2 - 6x + k = 0$ has equal roots?

A 6

B 3

C 9

D 12



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Quiz.

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Q. What is the value of 'k' if the quadratic equation $x^2 - 6x + k = 0$ has equal roots?

Solution

$$x^2 - 6x + k = 0$$

For equal roots

$$D = b^2 - 4ac = 0$$

$$36 - 4(1)k = 0$$

$$36 = 4k$$

$$k = 9$$



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3. Nature of Roots

Quiz.



Quiz.

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$4^2 + p(4) + 12 = 0$ $\quad p = -7$

Q. If one root of equation $x^2 + px + 12 = 0$ is 4, while equation $x^2 + px + q = 0$ has equal roots, then value of q

A $\frac{49}{4}$ B $\frac{4}{49}$ C 4 D 49

$x^2 - 7x + q = 0$

$D = (-7)^2 - 4(1)q = 0$

$49 = 4q$

$q =$

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Quiz.

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Q. If one root of equation $x^2 + px + 12 = 0$ is 4, while equation $x^2 + px + q = 0$ has equal roots, then value of q

A $\frac{49}{4}$

B $\frac{4}{49}$

C 4

D 49

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Q. If one root of equation $x^2 + px + 12 = 0$ is 4, while equation $x^2 + px + q = 0$ has equal roots, then value of q

Solution

$$x^2 + px + 12 = 0$$

$$16 + 4p + 12 = 0$$

$$p = -7$$

$\therefore$ 2nd equation becomes $x^2 - 7x + q = 0$

By the given condition $D = 0$

$$(-7)^2 - 4(1)(q) = 0$$

$$49 - 4q = 0$$

$$q = \frac{49}{4}$$

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$D_{(1)} = 1 - 4(a)(b) \geq 0$ $\quad 4ab < 1$

Q. If the roots of the equation $ax^2 + x + b = 0$ be real, and unequal then the roots of equation $x^2 - 4\sqrt{ab}x + 1 = 0$ will be

A Rational

B Irrational

C Real

D Imaginary

$4ab - 1 < 0$

$D = 16ab - 4 = 4(4ab - 1)$

$4(<0)$

$= <0$



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Quiz.

Chat On

Q. If the roots of the equation $ax^2+x+b=0$ be real, and unequal then the roots of equation $x^2-4\sqrt{ab}x+1=0$ will be

A Rational

B Irrational

C Real

D Imaginary

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Let's Solve

$\sqrt{3}+1$, $\sqrt{3}-1$ $\quad x^2-(\alpha+\beta)x+\alpha\beta=0$

Ex. (a) Form a quadratic equation with rational coefficient having $\sqrt{3}+1$ as one of the roots. → $\sqrt{3}-1$ (β) $1-\sqrt{3}$

(b) Evaluate value of x^3+3x^2+2 at $x=1+\sqrt{3}$

$ax^2+bx+c=0$

$D<0$

$a,b,c \in \mathbb{R}$

$p+iq, p-iq$

$D>0$

D is not Perfect Sqr

$a,b,c \in \mathbb{Q}$

$p+\sqrt{q}, p-\sqrt{q}$



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Let's Solve

Ex. (a) Form a quadratic equation with rational coefficient having $\sqrt{3}+1$ as one of the roots.

(b) Evaluate value of x^3+3x^2+2 at $x=1+\sqrt{3}$

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Let's Solve

Ex. (a) Form a quadratic equation with rational coefficient having $\sqrt{3}+1$ as one of the roots.

$\sqrt{3}-1$ & $1-\sqrt{3}$

Solution

Since quadratic equation has rational coefficients and $\sqrt{3}+1$ is one of the root , thus other root is $-\sqrt{3}+1$

Thus required equation will be

$$x^2-\left(\sqrt{3}+1-\sqrt{3}+1\right)x+\left(\sqrt{3}+1\right)\left(-\sqrt{3}+1\right)=0$$

$\alpha + \beta$ $\quad$ $\alpha\beta$

$$\Rightarrow x^2-2x-2=0$$

$x^2 - S_1 x + S_2 = 0$

$x^2 - Sx + P = 0$

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Let's Solve

Ex. (a) Form a quadratic equation with rational coefficient having $\sqrt{3}+1$ as one of the roots.

Solution

Since quadratic equation has rational coefficients and $\sqrt{3}+1$ is one of the root , thus other root is $-\sqrt{3}+1$

Thus required equation will be

$$x^2-\left(\sqrt{3}+1-\sqrt{3}+1\right)x+\left(\sqrt{3}+1\right)\left(-\sqrt{3}+1\right)=0$$

$$\Rightarrow x^2-2x-2=0$$

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Let's Solve

Ex. (b) Evaluate value of x^3+3x^2+2 at $x=1+\sqrt{3}$

Solution

Given $x=1+\sqrt{3}$

or $x-1=\sqrt{3}$

Squaring both sides

$(x-1)^2=3$

$x^2+1-2x=3$

$x^2-2x-2=0$

Now to get value of

x^3+3x^2+2

$$
\begin{array}{r}
x+5 \\
x^2-2x-2 \overline{)\,x^3+3x^2+2} \\
-x^3+2x^2-2x \\
\hline
5x^2+2x+2 \\
-5x^2-10x-10 \\
\hline
12x+12
\end{array}
$$

$$x^3+3x^2+2=(x^2-2x-2)(x+5)+12(x+1)$$

$$=0+12(x+1)$$

$$=12(1+\sqrt{3}+1)$$

$$=12(2+\sqrt{3})$$

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Quadratic Equations

Remark

If $ax^2 + bx + c = 0$ is an identity,
that is, $ax^2 + bx + c = 0$ is valid for all real values of x,
or, $ax^2 + bx + c = 0$ has more than two real roots,
then, $a = b = c = 0$

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Let's Solve



Let's Solve

Ex. Find 't' such that $(t^2-3t+2)x^2+(t^2-5t+6)x+(t^2-4)=0$ is an identity.

→ 2 Values

$= 0$ $= 0$ $= 0$

more than 2

$t = 2$

$t^2-3t+2=0$

$(t-1)(t-2)=0$

$t = 1, 2$

$t^2-5t+6=0$

$(t-2)(t-3)=0$

$t = 2, 3$

$t^2-4=0$

$t = \pm 2$

$\cancel{-2}$ 2

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Let's Solve

Ex. Find 't' such that $\left(t^2-3t+2\right)x^2+\left(t^2-5t+6\right)x+\left(t^2-4\right)=0$ is an identity.

Solution

Given, $\left(t^2-3t+2\right)x^2+\left(t^2-5t+6\right)x+\left(t^2-4\right)=0$ is an identity.

So, coefficients of all terms should be zero simultaneously.

Thus,

$t^2-3t+2=0$ & $t^2-5t+6=0$ & $t^2-4=0$

$(t-2)(t-1)=0$ $(t-2)(t-3)=0$ $t=\pm 2$

$t=1,2$ $t=2,3$

Thus $t=2$.

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Quiz.



Quiz.

Chat Off

$9k(k-2)+10(k-2)=0$

$k=-\frac{10}{9}$ or 2

Q. For what values of k will the equation $x^2-2(1+3k)x+7(3+2k)=0$ have equal roots

$D=0$

A $1, -\frac{10}{9}$ B $2, -\frac{10}{9}$ C $3, -\frac{10}{9}$ D $4, -\frac{10}{9}$

$D=4(1+3k)^2-4(1)7(3+2k)=0$

$1+9k^2+6k-21-14k=0$

$9k^2-8k-20=0$

$9k^2-18k+10k-20=0$

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Quiz.

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Q. For what values of k will the equation $x^2 - 2(1+3k)x + 7(3+2k) = 0$ have equal roots

A $1, -\frac{10}{9}$ **B** $2, -\frac{10}{9}$ **C** $3, -\frac{10}{9}$ **D** $4, -\frac{10}{9}$

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Quiz.

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Q. For what values of k will the equation $x^2 - 2(1+3k)x + 7(3+2k) = 0$ have equal roots

Solution

Since roots are equal then $\left[-2(1+3k)\right]^2 - 4.1.7(3+2k) = 0$

$\Rightarrow 1 + 9k^2 + 6k = 21 + 14k$

$\Rightarrow 9k^2 - 8k - 20 = 0$

Solving, we get $k = 2, -\frac{10}{9}$

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Summary.



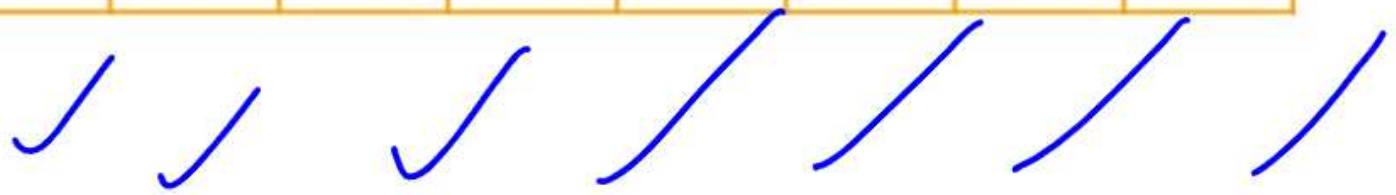
Summary.

1. Relation between roots and coefficients
2. Nature of Roots

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Tatva Ex-1 HW Q. No's

Q. No's.							
7	8	9	10	11	12	45	46

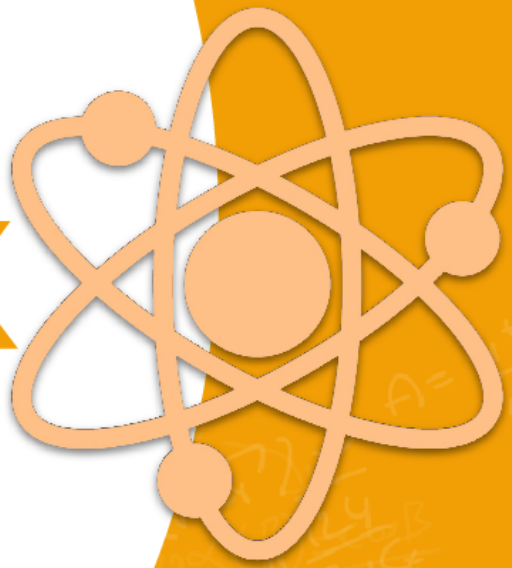


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Thanks.



Feedback



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Quadratic Equations

Grade 11 • Mathematics • Module 3



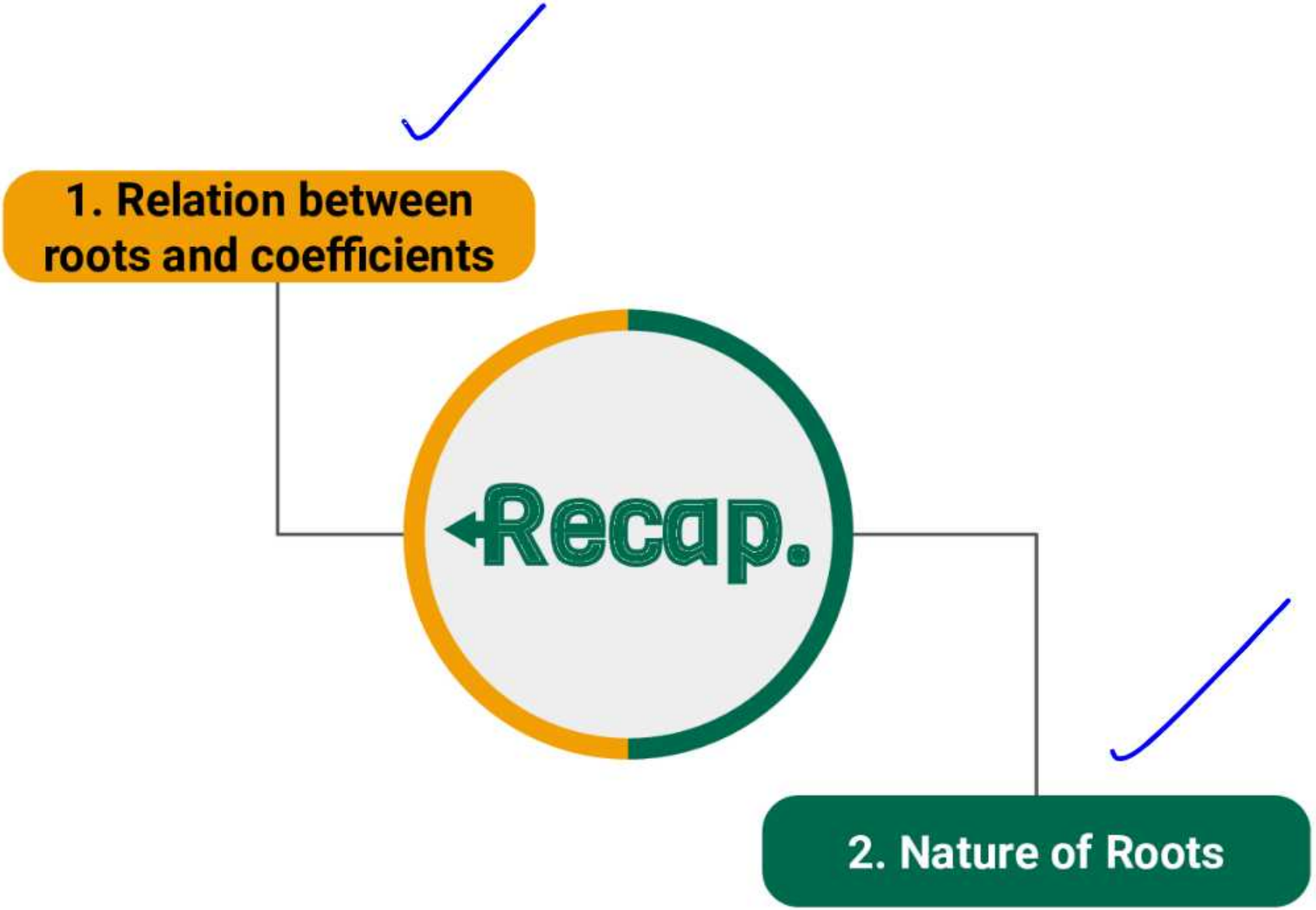




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Recap.





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Quiz.



Quiz.

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$D = (17)^2 - 4(8)(P)$

Q. If the equation $8x^2 + 17x + P = 0$ has equal roots then the then the value of P is

A $\frac{280}{32}$

B $\frac{289}{2}$

C $\frac{289}{32}$

D $\frac{1}{32}$

$(17)^2 - 32P = 0$

$P = \frac{289}{32}$

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Quiz.

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Q. If the equation $8x^2 + 17x + P = 0$ has equal roots then the then the value of P is

A $\frac{280}{32}$

B $\frac{289}{2}$

C $\frac{289}{32}$

D $\frac{1}{32}$



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Q. If the equation $8x^2 + 17x + P = 0$ has equal roots then the then the value of P is

Solution

Given $8x^2 + 17x + P = 0$

For equal roots

$$D = b^2 - 4ac = 0$$

$$(17)^2 - 4(8)(P) = 0$$

$$P = \frac{289}{32}$$

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Contents

1. Polynomial
2. Relation between Roots and Coefficients
3. Nature of roots
4. **Transformation of Equations**
5. **Common Roots**

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1. Polynomial
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4. Transformation of Equation

$f(x) \equiv ax^2 + bx + c \equiv 0$

α, β

- $\alpha+k, \beta+k$ — roots → $f(x-k) \equiv a(x-k)^2 + b(x-k) + c = 0$
- $\alpha-k, \beta-k$ — roots → $f(x+k) \equiv a(x+k)^2 + b(x+k) + c = 0$
- $k\alpha, k\beta$ — roots → $f\left(\frac{x}{k}\right) = 0$
- $\frac{\alpha}{k}, \frac{\beta}{k}$ — roots → $f(kx) = 0$
- α^n, β^n — roots → $f\left(x^{1/n}\right) = 0$

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$\alpha^{1/n}, \beta^{1/n}$ roots $f(x^n) = 0$

$\frac{1}{\alpha}, \frac{1}{\beta}$ roots $f\left(\frac{1}{x}\right) = 0$

$-\alpha, -\beta$ roots $f(-x) = 0$

4. Transformation of Equation

If roots of $ax^2+bx+c=0$ are α & β then equation having roots:

(a) 2α & 2β is → $f\left(\frac{x}{2}\right)=a\left(\frac{x}{2}\right)^2+b\left(\frac{x}{2}\right)+c=0$

(b) $\frac{1}{\alpha}$ & $\frac{1}{\beta}$ is → $f\left(\frac{1}{x}\right)=a\left(\frac{1}{x}\right)^2+b\left(\frac{1}{x}\right)+c=0$

(c) $\alpha+2$ & $\beta+2$ is → $f(x-2)=a(x-2)^2+b(x-2)+c=0$

(d) α^2 & β^2 is → $f\left(x^{1/2}\right)=a\left(x^{1/2}\right)^2+b\left(x^{1/2}\right)+c=0$

Hint

This shortcut works for equation of any degree.

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Let's Solve



Let's Solve

Ex. The equation whose roots are negative of the roots of the equation $x^3 - 3x^2 + x + 1 = 0$

$f(x) \equiv$ old

$\hookrightarrow \alpha, \beta, \gamma$ old ✓

$\downarrow$

$-\alpha, -\beta, -\gamma$ new roots

$\downarrow$ new Eqn

$f(-x) \equiv (-x)^3 - 3(-x)^2 + (-x) + 1 = 0$

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Let's Solve

Ex. The equation whose roots are negative of the roots of the equation $x^3 - 3x^2 + x + 1 = 0$

Solution

Given, $x^3 - 3x^2 + x + 1 = 0$

Replace $x \to -x$

$$-x^3 - 3x^2 - x + 1 = 0$$

$$x^3 + 3x^2 + x - 1 = 0$$

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Let's Solve

Ex. If α, β are the roots of $x^2+2x+2=0$ the find the equation whose roots are $\alpha+3$ and $\beta+3$

A $x^2-4x-5=0$

B $x^2-4x+5=0$

C $x^2+4x+5=0$

D $x^2+4x-5=0$

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Let's Solve

Ex. If α, β are the roots of $x^2+2x+2=0$ the find the equation whose roots are $\alpha+3$ and $\beta+3$

A $x^2-4x-5=0$

B $x^2-4x+5=0$

C $x^2+4x+5=0$

D $x^2+4x \quad 5=0$

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Let's Solve

Ex. If α, β are the roots of $x^2+2x+2=0$ the find the equation whose roots are $\alpha+3$ and $\beta+3$

Solution

To find the equation of $\alpha+3$ and $\beta+3$

Let's assume $\alpha+3=y \Rightarrow \alpha=y-3$

Now α is root of $x^2+2x+2=0$

$\therefore\ \alpha^2+2\alpha+2=0$

$\Rightarrow (y-3)^2+2(y-3)+2=0$

$y^2-6y+9+2y-6+2=0$

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Let's Solve

Ex. If α, β are the roots of $x^2+2x+2=0$ the find the equation whose roots are $\alpha+3$ and $\beta+3$

Solution

$$y^2-4y+5=0$$

It can be written in terms of x is

$$x^2-4x+5=0$$

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Quiz.



Quiz.

Chat Off

Q. The equation whose roots are reciprocal of the roots of the equation $3x^3 - 20x + 17 = 0$ is

A $3x^2 + 20x - 17 = 0$

B $17x^2 - 20x + 3 = 0$

C $17x^2 + 20x + 3 = 0$

D None of these



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Quiz.

Chat On

Q. The equation whose roots are reciprocal of the roots of the equation $3x^2 - 20x + 17 = 0$ is

A $3x^2 + 20x - 17 = 0$

B $17x^2 - 20x + 3 = 0$

C $17x^2 + 20x + 3 = 0$

D None of these



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Quiz.

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Q. The equation whose roots are reciprocal of the roots of the equation $3x^2 - 20x + 17 = 0$ is

Solution

Given, $3x^2 - 20x + 17 = 0$

Replace $x \to \frac{1}{x}$, we get

$$\frac{3}{x^2} - \frac{20}{x} + 17 = 0$$

$$\Rightarrow 3 - 20x + 17x^2 = 0 \Rightarrow 17x^2 - 20x + 3 = 0$$



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Let's Solve



Let's Solve

Ex. If α, β are the roots of the equation $x^2 - 3x + 7 = 0$. Find the equation of whose roots are $\frac{1}{\alpha + 1}$ and $\frac{1}{\beta + 1}$

A $x^2 - 5x + 1 = 0$

B $5x^2 - 11x + 1 = 0$

C $11x^2 - 5x + 1 = 0$

D $11x^2 + 5x - 1 = 0$

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Let's Solve

Ex. If α, β are the roots of the equation $x^2 - 3x + 7 = 0$. Find the equation of whose roots are $\frac{1}{\alpha + 1}$ and $\frac{1}{\beta + 1}$

A $x^2 - 5x + 1 = 0$

B $5x^2 - 11x + 1 = 0$

C $11x^2 - 5x + 1 = 0$

D $11x^2 + 5x - 1 = 0$

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Let's Solve

Ex. If α, β are the roots of the equation $x^2 - 3x + 7 = 0$. Find the equation of whose roots are $\frac{1}{\alpha + 1}$ and $\frac{1}{\beta + 1}$

Solution

To find the equation of $\frac{1}{\alpha + 1}$ and $\frac{1}{\beta + 1}$

Let's assume $\frac{1}{\alpha + 1} = y$

$$\Rightarrow \frac{1}{y} = \alpha + 1$$

$$\Rightarrow \alpha = \frac{1}{y} - 1 = \frac{1 - y}{y}$$

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Let's Solve

Ex. If α, β are the roots of the equation $x^2 - 3x + 7 = 0$. Find the equation of whose roots are $\frac{1}{\alpha + 1}$ and $\frac{1}{\beta + 1}$

Solution

α is root of $x^2 - 3x + 7 = 0$

$\therefore \ \alpha^2 - 3\alpha + 7 = 0$

$$\left(\frac{1-y}{y}\right)^2 - 3\left(\frac{1-y}{y}\right) + 7 = 0$$

$$(1-y)^2 - 3(1-y)y + 7y^2 = 0$$

$$1 - 2y + y^2 - 3y + 3y^2 + 7y^2 = 0$$

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Let's Solve

Ex. If α, β are the roots of the equation $x^2 - 3x + 7 = 0$. Find the equation of whose roots are $\frac{1}{\alpha + 1}$ and $\frac{1}{\beta + 1}$

Solution

$11y^2 - 5y + 1 = 0$

This can be written in terms of x as

$11x^2 - 5x + 1 = 0$



Did you understand everything till now? Poll

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Let's Solve

Ex. If α, β are the roots of the equation $ax^2 + bx + c = 0$, the find the roots of the equation $ax^2 - bx(x-1) + c(x-1)^2 + c(x-1)^2 = 0$ in terms of α and β.

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Let's Solve

Ex. If α, β are the roots of the equation $ax^2 + bx + c = 0$, the find the roots of the equation $ax^2 - bx(x-1) + c(x-1)^2 + c(x-1)^2 = 0$ in terms of α and β.

Solution

$$ax^2 - bx(x-1) + c(x-1)^2 + c(x-1)^2 = 0$$

$$\rightarrow \frac{ax^2}{(1-x)^2} + \frac{bx}{1-x} + c = 0 \qquad \ldots(1)$$

Now, α is a roots of $ax^2 + bx + c = 0$.

Then let, $\alpha = \frac{x}{1-x} \Rightarrow x = \frac{\alpha}{\alpha + 1}$

Hence, the roots of (1) are $\frac{\alpha}{(1+\alpha)}, \frac{\beta}{(1+\beta)}$

Quiz.



Quiz.

Chat Off

Q. If α, β are roots of $x^2 - 3x + 1 = 0$, then the equation whose roots are $\frac{1}{\alpha - 2}, \frac{1}{\beta - 2}$ is

A $x^2 + x - 1 = 0$

B $x^2 + x + 1 = 0$

C $x^2 - x - 1 = 0$

D None of these

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Quiz.

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Q. If α, β are roots of $x^2 - 3x + 1 = 0$, then the equation whose roots are $\frac{1}{\alpha - 2}, \frac{1}{\beta - 2}$ is

A $x^2 + x - 1 = 0$

B $x^2 + x + 1 = 0$

C $x^2 - x - 1 = 0$

D None of these



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Quiz.

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Q. If α, β are roots of $x^2 - 3x + 1 = 0$, then the equation whose roots are $\frac{1}{\alpha - 2}, \frac{1}{\beta - 2}$ is

Solution

$$\frac{1}{\alpha - 2} = y$$

$$\alpha - 2 = \frac{1}{y} \qquad \alpha = \frac{1}{y} + 2 \Rightarrow \frac{2y+1}{y}$$

Now, α is roots of $x^2 - 3x + 1 = 0$

$$\Rightarrow \alpha^2 - 3\alpha + 1 = 0$$

$$\frac{(2y+1)^2}{y^2} - \frac{3(2y+1)}{y} + 1 = 0$$

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Quiz.

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Q. If α, β are roots of $x^2 - 3x + 1 = 0$, then the equation whose roots are $\frac{1}{\alpha - 2}, \frac{1}{\beta - 2}$ is

Solution

$$4y^2 + 4y + 1 - 6y^2 - 3y + y^2 = 0$$

$$-y^2 + y + 1 = 0$$

$$\Rightarrow y^2 - y - 1 = 0$$

$\Rightarrow$ In terms of x we can write $x^2 - x - 1 = 0$



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6 Analysis of graph of Quadratic Equations

7 Location of Roots

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5. Common Roots

5. Common Roots

Given two Quadratic Equations, there are two possibilities-

1. They have one root in common

2. Both the roots are common

Let's first learn how to handle questions of one root common

5. Common Roots

Method-I

When we know both roots of one equation make them satisfy second equation one by one.

Let's Solve



Let's Solve

Ex. Find 'K' such that $2x^2 + Kx - 5 = 0$ & $x^2 - 3x - 4 = 0$ may have a root in common

Let's Solve

Ex. Find 'K' such that $2x^2 + Kx - 5 = 0$ & $x^2 - 3x - 4 = 0$ may have a root in common

Solution Given, $2x^2 + Kx - 5 = 0$ and $x^2 - 3x - 4 = 0$ have a common root.

$x^2 - 3x - 4 = 0$

$x^2 - 4x + x - 4 = 0$

$(x-4)(x+1) = 0$

$x = 4, -1$

Now, if 4 is the common root

Then, $2(4)^2 + 4K - 5 = 0$

$4K = -27$

$K = \frac{-27}{4}$

if -1 is the common root

Then,

$2(-1)^2 - K - 5 = 0$

$k = -3$

So $k = -3$ or $\frac{-27}{4}$

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Quiz.



Quiz.

Chat Off

Q. Find k such that $x^2 + kx - 3 = 0$ and $x^2 - 4x + 3 = 0$ may have one root in common

A -1 or -2 **B** -2 or 2 **C** 1 or -2 **D** 1 or 2



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Quiz.

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Q. Find k such that $x^2 + kx - 3 = 0$ and $x^2 - 4x + 3 = 0$ may have one root in common

A -1 or -2

B -2 or 2

C 1 or -2

D 1 or 2

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Quiz.

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Q. Find k such that $x^2 + kx - 3 = 0$ and $x^2 - 4x + 3 = 0$ may have one root in common

Solution

$$x^2 - 4x + 3 = 0$$

$$(x-3)(x-1) = 0$$

$$x = 1, x = 3$$

Now if $x = 1$ is a common roots then

$$1^2 + k(1) - 3 = 0$$

$$k = 2$$

If $x = 3$ is a common root then

$$3^2 + 3(k) - 3 = 0$$

So, $k =$ -2 or 2

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5. Common Roots

Method-II

Solve two equations to get common root

Let's Solve



Let's Solve

Ex. If $x^2+bx+c=0$ and $x^2+cx+b=0$ have a common root then find their uncommon root. Given $b \neq c$

Let's Solve

Ex. If $x^2+bx+c=0$ and $x^2+cx+b=0$ have a common root then find their uncommon root. Given $b \neq c$

Solution

Given $x^2+bx+c=0$ and $x^2+cx+b=0$ have a common root

Let it be α. Thus

$\alpha^2+b\alpha+c=0$ and $\alpha^2+c\alpha+b=0$

$\alpha(b-c)=(b-c)$

$\alpha=1$

Let other roots be β and β'

Now $\alpha\cdot\beta=c \quad \alpha\cdot\beta'=b$

$\beta=c \quad \beta'=b$

Also,

$\alpha+\beta=-b \quad \& \quad \alpha+\beta'=-c$

$\therefore \beta=-b-1 \quad \& \quad \therefore \beta'=-c-1$

Let's Solve

Ex. If the equations $x^2+ax+12=0$, $x^2+bx+15=0$ and $x^2+(a+b)x+36=0$ have a common positive root, then find the values of a and b.

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Let's Solve

Ex. If the equations $x^2+ax+12=0$, $x^2+bx+15=0$ and $x^2+(a+b)x+36=0$ have a common positive root, then find the values of a and b.

Solution

We have,

$$x^2+ax+12=0 \quad ...(1)$$

$$x^2+bx+15=0 \quad ...(2)$$

Adding (1) and (2) we get

$$2x^2+(a+b)x+27=0$$

Now subtracting it from the third given equation, we get

$$x^2-9=0 \Rightarrow x=3,-3$$

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Let's Solve

Ex. If the equations $x^2+ax+12=0$, $x^2+bx+15=0$ and $x^2+(a+b)x+36=0$ have a common positive root, then find the values of a and b.

Solution

Thus, common positive root is 3. Hence,

$$9+12+3a=0$$

$$\Rightarrow a=-7 \text{ and } 9+3b+15=0$$

$$\Rightarrow b=-8$$

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5. Common Roots

Method-III

Applying proper condition

Condition for one root common in

$a_1x^2+b_1x+c_1=0$

$a_2x^2+b_2x+c_2=0$

is $(c_1a_2-c_2a_1)^2=(b_1c_2-b_2c_1)(a_1b_2-a_2b_1)$

Let's Solve



Let's Solve

Ex. If $ax^2+bx+c=0$ & $bx^2+cx+a=0$ have a common root then find relation between a, b & c



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Let's Solve

Ex. If $ax^2+bx+c=0$ & $bx^2+cx+a=0$ have a common root then find relation between a, b & c

Solution

Let α be the common root

So, $a\alpha^2+b\alpha+c=0$

$b\alpha^2+c\alpha+a=0$

Applying condition for common root

$$(c_1a_2-c_2a_1)^2=(b_1c_2-c_1b_2)(a_1b_2-b_1a_2)$$

$$\therefore \quad (bc-a^2)^2=(ab-c^2)(ac-b^2)$$

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Let's Solve

Ex. If $ax^2+bx+c=0$ & $bx^2+cx+a=0$ have a common root then find relation between a, b & c

Solution

$$b^2c^2+a^4-2a^2bc=\left(ab-c^2\right)\left(ac-b^2\right)$$

$$a^3+b^3+c^3=3abc$$

Thus, either $a+b+c=0$ or $a=b=c$

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Let's Solve

Ex. If a root of the equations $x^2 + px + q = 0$ and $x^2 + \alpha x + \beta = 0$ is common, then its value will be (where $p \neq \alpha$ and $q \neq \beta$)

A $\dfrac{q-\beta}{\alpha-p}$

B $\dfrac{p\beta-\alpha q}{q-\beta}$

C $\dfrac{q-\beta}{\alpha-p}$ or $\dfrac{p\beta-\alpha q}{q-\beta}$

D None of these

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Let's Solve

Ex. If a root of the equations $x^2 + px + q = 0$ and $x^2 + \alpha x + \beta = 0$ is common, then its value will be (where $p \neq \alpha$ and $q \neq \beta$)

A $\dfrac{q-\beta}{\alpha-p}$

B $\dfrac{p\beta-\alpha q}{q-\beta}$

C $\dfrac{q-\beta}{\alpha-p}$ or $\dfrac{p\beta-\alpha q}{q-\beta}$

D None of these

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Let's Solve

Ex. If a root of the equations $x^2 + px + q = 0$ and $x^2 + \alpha x + \beta = 0$ is common, then its value will be (where $p \neq \alpha$ and $q \neq \beta$)

Solution

Let the common root be y. Then $y^2 + py + q = 0$ and $y^2 + \alpha y + \beta = 0$

On solving by cross multiplication, we have

$$\frac{y^2}{p\beta - q\alpha} = \frac{y}{q - \beta} = \frac{1}{\alpha - p}$$

$$\therefore y = \frac{q - \beta}{\alpha - p} \quad \text{and} \quad \frac{y^2}{y} = y = \frac{p\beta - q\alpha}{q - \beta}$$

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Quiz.



Quiz.

Chat Off

Q. If the two equations $x^2 - cx + d = 0$ and $x^2 - ax + b = 0$ have one common root and the second has equal roots, then $2(b+d) =$

A 0

B $a+c$

C ac

D $-ac$



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Quiz.

Chat On

Q. If the two equations $x^2 - cx + d = 0$ and $x^2 - ax + b = 0$ have one common root and the second has equal roots, then $2(b+d) =$

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Quiz.

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Q. If the two equations $x^2 - cx + d = 0$ and $x^2 - ax + b = 0$ have one common root and the second has equal roots, then $2(b+d) =$

Solution

Let roots of $x^2 - cx + d = 0$ be α, β then roots of $x^2 - ax + b = 0$ be α, α

$\therefore \alpha + \beta = c, \alpha\beta = d, \alpha + \alpha = a, \alpha^2 = b$

Hence $2(b+d) = 2(\alpha^2 + \alpha\beta) = 2\alpha(\alpha + \beta) = ac$



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Q. If $x^2 - hx - 21 = 0, x^2 - 3hx + 35 = 0$ $(h > 0)$ has a common root, then the value of h is equal to

A 1 **B** 2 **C** 3 **D** 4



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Q. If $x^2 - hx - 21 = 0, x^2 - 3hx + 35 = 0$ $(h > 0)$ has a common root, then the value of h is equal to

A 1

B 2

C 3

D 4



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Q. If $x^2 - hx - 21 = 0, x^2 - 3hx + 35 = 0$ $(h > 0)$ has a common root, then the value of h is equal to

Solution

Subtracting, we get $2hx = 56$ or $hx = 28$

Putting in any, $x^2 = 49$

$$\therefore \left[\frac{28}{h}\right]^2 = 7^2 \Rightarrow h = 4(h > 0)$$



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Let's Solve



Let's Solve

Ex. $x^2 - 11x + a$ and $x^2 - 14x + 2a$ will have a common factor, if $a =$

(A) 24 (B) 0, 24 (C) 3, 24 (D) 0, 3

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Let's Solve

Ex. $x^2 - 11x + a$ and $x^2 - 14x + 2a$ will have a common factor, if $a =$

A 24

B 0, 24

C 3, 24

D 0, 3

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Let's Solve

Ex. $x^2-11x+a$ and $x^2-14x+2a$ will have a common factor, if $a=$

Solution

Expressions are $x^2-11x+a$ and $x^2-14x+2a$ will have a common factor, then

$$\Rightarrow \frac{x^2}{-22a+14a}=\frac{x}{a-2a}=\frac{1}{-14+11}$$

$$\Rightarrow \frac{x^2}{-8a}=\frac{x}{-a}=\frac{1}{-3} \Rightarrow x^2=\frac{8a}{3} \text{ and } x=\frac{a}{3}$$

$$\Rightarrow \left(\frac{a}{3}\right)^2=\frac{8a}{3} \Rightarrow \frac{a^2}{9}=\frac{8a}{3} \Rightarrow a=0,\ 24$$

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Quiz.



Quiz.

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Q. If the equation $x^2 + ax + b = 0$ and $x^2 + bx + a =$ have a common root, then their other roots satisfy the equation.

A $x^2 + x + ab = 0$

B $x^2 - x + ab = 0$

C $x^2 - x - ab = 0$

D $x^2 + x - ab = 0$



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Quiz.

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Q. If the equation $x^2+ax+b=0$ and $x^2+bx+a=$ have a common root, then their other roots satisfy the equation.

A $x^2+x+ab=0$

B $x^2-x+ab=0$

C $x^2-x-ab=0$

D $x^2+x-ab=0$



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Quiz.

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Q. If the equation $x^2+ax+b=0$ and $x^2+bx+a=$ have a common root, then their other roots satisfy the equation.

Solution

'1' is a common root

$\therefore a+b=-1$

other roots are b,a

$x^2-x(a+b)+ab-0$

$x^2-x(-1)+ab=0$



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Summary.



Summary.

1. Transformation of Equation
2. Common Roots

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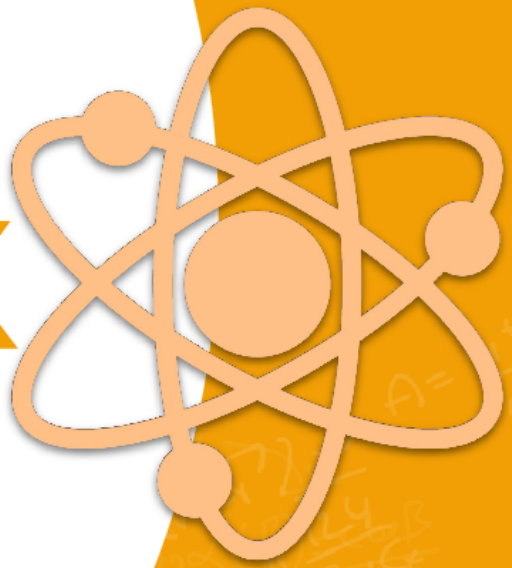
Tatva Ex-1 HW Q. No's

Q. No's.		
26	27	28

Thanks.



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